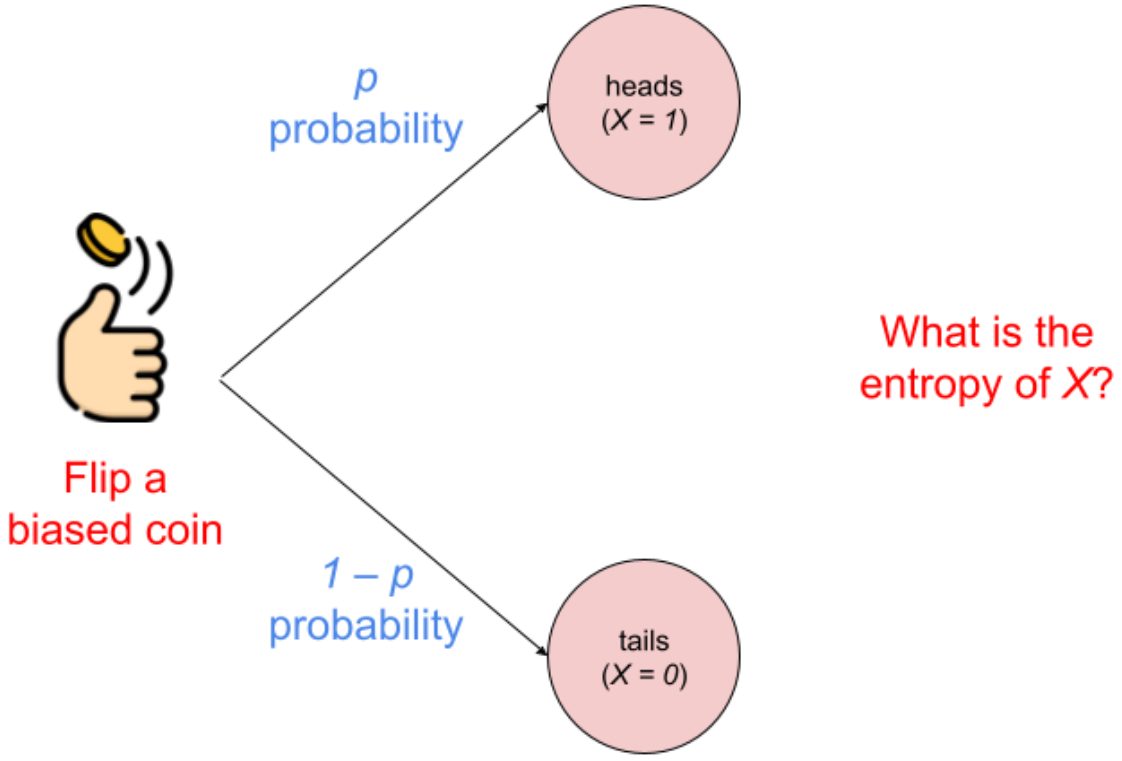


Here, the entropy of X is defined to be $-p \log p - (1 - p) \log(1 - p)$, where *log* is the logarithmic function in base 2. We see that this is a function of the probability of heads p. Figure 6.1 shows the function of entropy for p in $(0, 1)$, from which we can gain some insights:

- The entropy is always non-negative.
- The entropy increases when p increases before 0.5, reaches its highest at $p = 0.5$, and then decreases afterwards.

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>